RoHS

LDMOS RF Power Transistor

HTU7G06S002P

100-1000MHz, 2W, 4V
WIDE BAND
RF POWER LDMOS TRANSISTOR

1. Product Description

HTU7G06S002P is an LDMOS RF power transistor designed for VHF/UHF band RF power amplifiers.

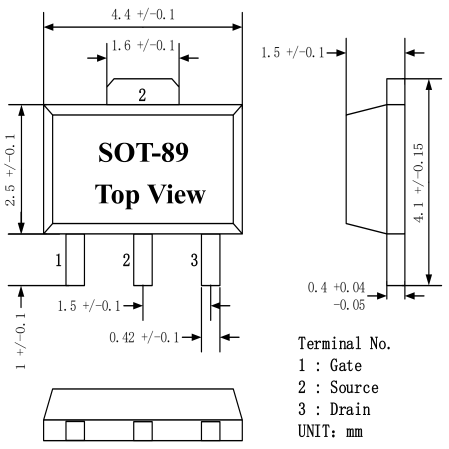
The device has an integrated electrostatic protection circuit.

2. Typical performance

Frequency (MHz)	VDD (V)	Pin (W)	Pout (W)	Eff (%)
400-470[1]	4.00	0.10	2.84	61.1

1. Based on 4V, UHF band, 2W reference design performance test

2. The above performance is the minimum performance of the reference design in the entire band



3. Product application

- VHF band handheld walkie-talkie
- UHF band handheld walkie-talkie
- 100-1000MHz other application driver or final amplifier

4. Absolute maximum ratings

Parameter	Symbol	Condition	Value	Unit
Drain-source voltage	V_{DSS}	V_{GS} = 0V	-0.5 ~ 12	V
Gate-source voltage	V_{GG}	V_{DS} = 0V	-5 ~ 10	V
Operating voltage	V_{DD}	-	6	V
Storage temperature	T_{stg}	-	-55 ~ 150	℃
Operating junction temperature	T_J	-	-40 ~ 150	℃
Thermal resistance	Zth(j-c)	-	10	℃/W

5. Electrical characteristics

Parameter	Symbol	Test Condition	Minimum	Typical	Maximum	Unit
Breakdown voltage	$V_{(BR)DDS}$	V_{GG}=0V, I_D=38.4uA	12	-	-	V
Drain leakage current	I_{DSS}	V_{DD}=12V, V_{GG}=0V	-	-	1	uA
Gate leakage current	I_{GSS}	V_{DD}=0V, V_{GG}=10V	-	-	1	uA
Threshold voltage	V_{th}	V_{DD}=V_{GG}, I_D=38.4uA	0.6	0.9	1.2	V
Output power	P_{out}	V_{DD}=4V, P_{in}=0.1W f=430MHz, I_{dq}=300mA	-	3.2	-	W
Drain efficiency	η_d		-	66	-	%

6. ESD characteristics

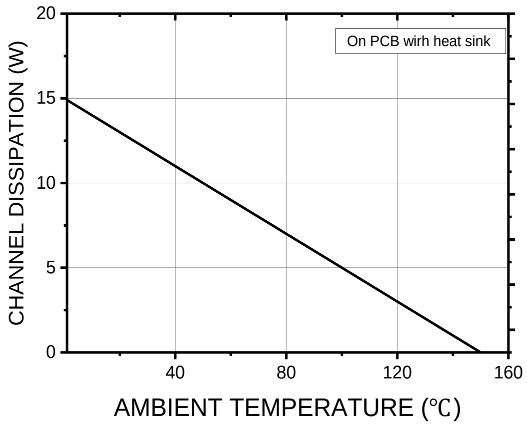
Condition	Grade
HBM (JESD22 - A114)	1C
MM (EIA/JESD22 - A115)	A
CDM (JESD22 - C101)	III

7. Load mismatch test (based on Huatai Demo PA test circuit)

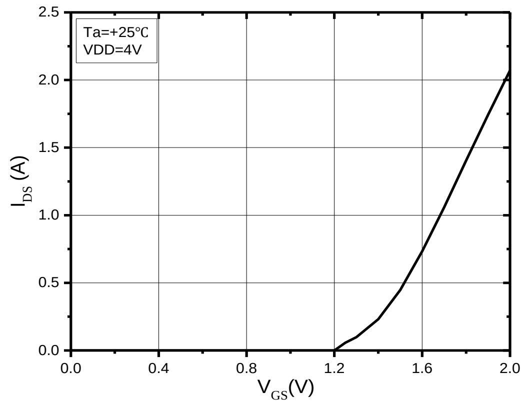
Condition	Result
VSWR = 20:1 at all Phase Angles CW: VDD=4.2V, IDQ=300mA, f=435MHz, Pout = 35dBm.	Transistor performance remains unchanged

8. Typical DC performance

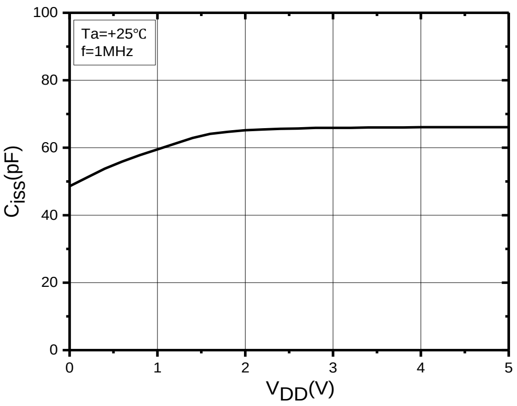
CHANNEL DISSIPATION VS.
AMBIENT TEMPERATURE



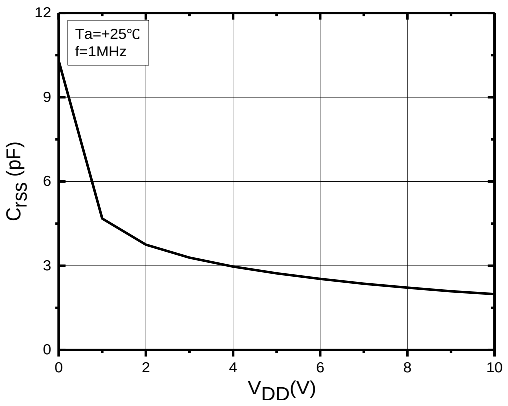
I_{DS} VS. V_{GS}



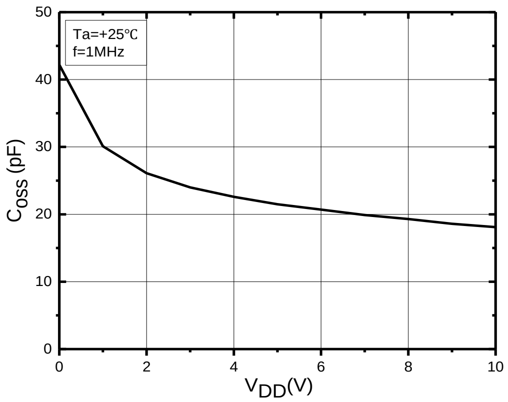
C_{iss} VS. V_{GS}



C_{rss} VS. V_{DS}

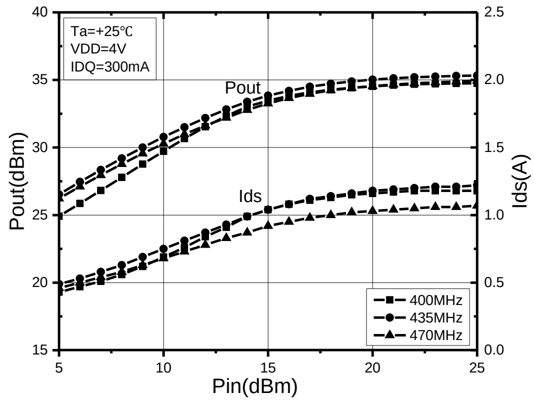


C_{oss} VS. V_{DS}

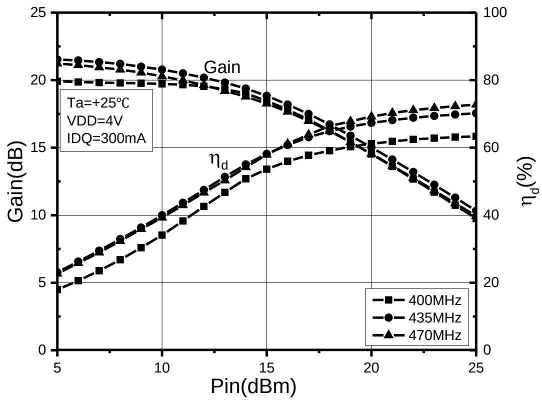


9. UHF-Band 2W typical performance (f=400-470MHz, VDD=4V)

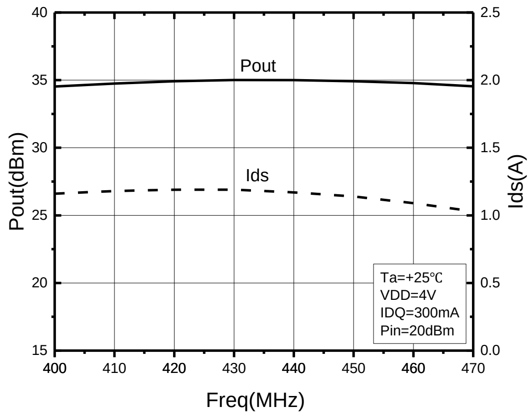
Pout,Ids VS. Pin



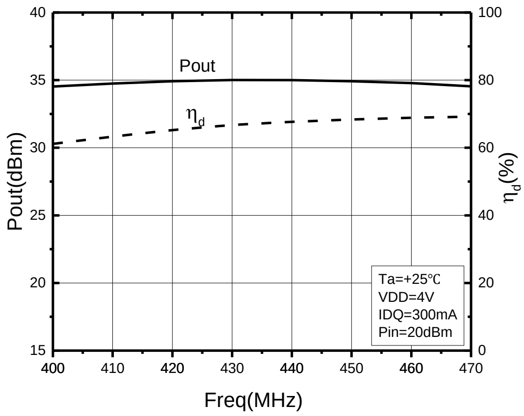
Gain,Eff VS. Pin



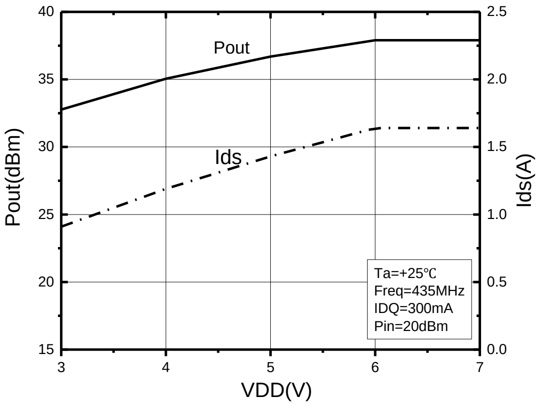
Pout,Ids VS. Freq



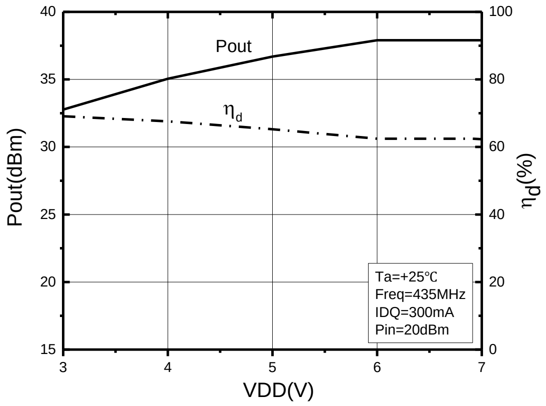
Pout,Eff VS. Freq



Pout,Ids VS. VDD

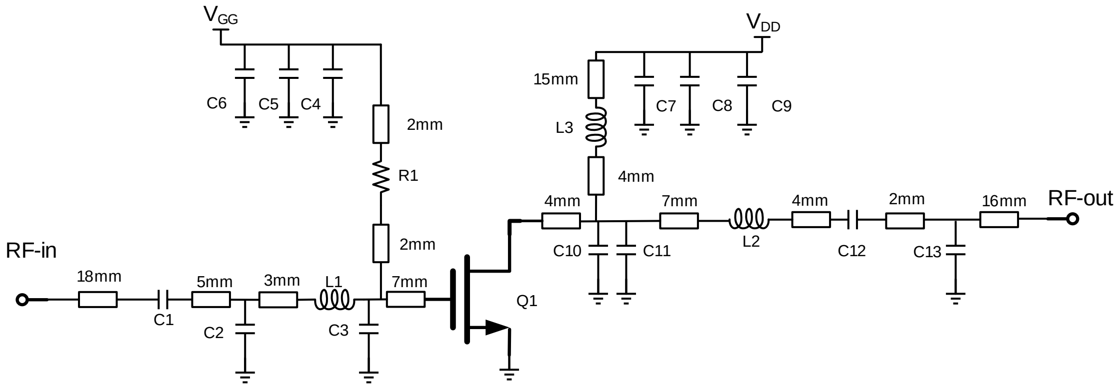


Pout,Eff VS. VDD



10. UHF-Band 4V 2W reference design

400-470MHz (@VDD = 4V, IDQ =300mA)

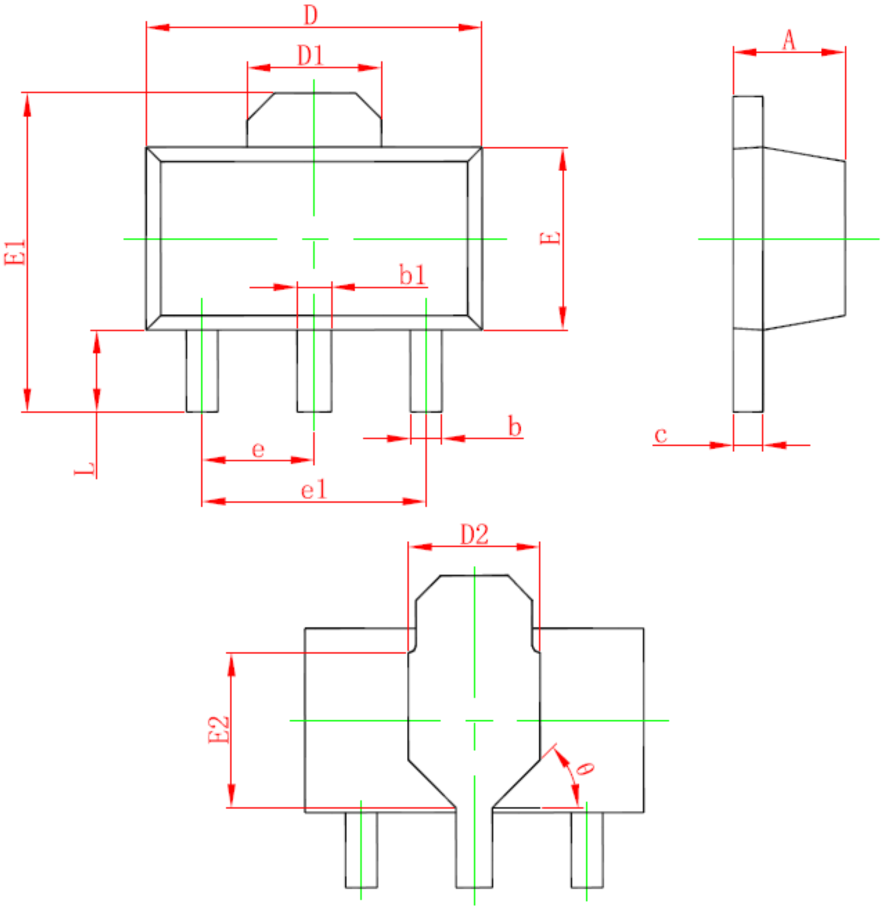


Note: The characteristic impedance of microstrip lines is 50 ohms.

Designator	Description	Model	Manufacturer
C4, C7, C12	100 pF Chip Capacitors	GRM1885C1H101JA01	muRata
C1	33pF Chip Capacitor	GRM1885C1H330JA01	muRata
C2	22 pF Chip Capacitor	GRM1885C1H220JA01	muRata
C3	12 pF Chip Capacitors	GRM1885C1H120JA01	muRata
C10	27 pF Chip Capacitor	GRM1885C1H270JA01	muRata
C11	18 pF Chip Capacitors	GRM1885C1H180JA01	muRata
C13	8 pF Chip Capacitors	GRM1885C1H8R0JA01	muRata
C6	4.7 uF Chip Capacitors	GRM32ER61H474KA12L	muRata
C5, C8	1 nF Chip Capacitors	GRM1885C1H102JA01	muRata
C9	10 uF Chip Capacitors	GRM32ER61H105KA12L	muRata
L1	1.2 nH Chip Inductor	0603	muRata
L2	W.D.: 0.40mm, I.D.: 1.5mm, 2 turns	-	-
L3	W.D.: 0.35mm, I.D.: 1.5mm, 8 turns	-	-
R1	51 Ω Chip Resistor	-	-
Q1	RF LDMOS	HTU7G06S002P	Suzhou Huatai Electronics Ltd.
PCB	εr = 4.3	FR4	-

11. Package size and pin distribution

SOT-89 package size and pin distribution



Symbol	Dimesions in Milimeters		Dimesions in Inches	
	Min.	Max.	Min.	Max.
A	1.400	1.600	0.055	0.063
b	0.320	0.520	0.013	0.020
b1	0.400	0.580	0.016	0.023
c	0.350	0.440	0.014	0.017
D	4.400	4.600	0.173	0.181
D1	1.550 REF.		0.061 REF.	
D2	1.750 REF.		0.069 REF.	
E	2.300	2.600	0.091	0.102
E1	3.940	4.250	0.155	0.167
E2	1.900 REF.		0.075 REF.	
e	1.500 TYP.		0.060 TYP.	
e1	3.000 TYP.		0.118 TYP.	
L	0.900	1.200	0.035	0.047
θ	45°		45°	